



Air University, Islamabad

**Software Requirement Specification
(SRS DOCUMENT)**

for

DailyMed

Version 1.0

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1 Introduction

The project DailyMed's Software Requirements Specification (SRS) describes a smart health follow-up platform designed to help patients with long-term conditions including diabetes and hypertension. Through intelligent warnings and predictive insights, the platform improves decision-making, decreases healthcare delays, and increases patient involvement by utilizing artificial intelligence and contemporary online technology. With its ability to monitor health vitals, facilitate communication between patients, caregivers, and doctors, and automate follow-up care procedures, DailyMed is a one-stop digital companion.goal.

1.1 Scope

The DailyMed system is an AI-powered, web-based platform that provides the following capabilities:

- Allows manual logging of vital signs like blood sugar and blood pressure.
- Utilizes machine learning models (LSTM for blood sugar forecasting, and SVM/Logistic Regression for hypertension risk analysis).
- Presents real-time visual feedback through an interactive dashboard.
- Sends automated alerts and notifications to patients, caregivers, and healthcare professionals in case of abnormal patterns or emergencies.
- The platform is entirely digital and does not include physical clinical services. It is focused on improving remote health monitoring and virtual care delivery.

1.2 Modules

1.2.1 Module 1: User Management Module

Purpose: Manage account creation, authentication, and role-based access.

Features:

- Sign up/login (Patients, Doctors, Caregivers)
- Password encryption & recovery
- Role-based dashboard redirection
- Profile update & deletion
- Secure session management

1.2.2 Module 2: Health Data Entry Module

Purpose: Collect and store health vitals manually entered by the user.

Features:

- Input fields for:

- o Blood Sugar (Fasting, Random, Post-meal)
- o Blood Pressure
- o Weight & Heart Rate
- Date-wise logging
- Reminder to enter data daily
- Option to attach reports or lab documents

1.2.3 Module 3: AI Prediction & Analytics Module

Purpose: Use ML models to analyze trends and provide health predictions.

Features:

- LSTM for blood sugar forecasting
- SVM/Logistic Regression for hypertension risk
- Custom threshold alerts
- Weekly health trend graphs
- Accuracy feedback loop for ML model training

1.2.4 Module 4: Insulin Dosage Adjustment Module

Purpose: Suggest changes to insulin dose based on glucose trends.

Features:

- AI-recommended insulin dosage (increase/decrease/maintain)
- Tracks insulin type and timing
- Doctor approval interface for final confirmation
- Explanation for each recommendation
- Alert system for critical insulin changes

1.2.5 Module 5: Alert & Notification System

Purpose: Keep users and their network informed about critical updates.

Features:

- Real-time alerts for abnormal vitals
- SMS/email/push notifications
- Medication reminders
- Emergency contact alerting
- Doctor/caregiver-specific alerts

1.2.6 Module 6: Data Visualization & Dashboard Module

Purpose: Present health data in an intuitive and easy-to-read format.

Features:

- Line graphs, pie charts, and bar graphs for:

- o Glucose trends
- o BP history
- o Medication adherence
- Risk indicator status (Low/Medium/High)
- Custom date filters

1.2.7 Module 7: Doctor/Professional Panel

Purpose: Provide doctors access to patient data and control features.

Features:

- View patients assigned to them
- Patient history & AI suggestions
- Approve/reject dosage recommendations
- Add digital prescriptions
- Schedule follow-ups or virtual appointments

1.2.8 Module 8: Caregiver Panel

Purpose: Allow caregivers to monitor their patient's health condition.

Features:

- Limited access to dashboards
- Emergency alerts only
- Read-only data visibility
- Option to message doctor or patient

1.2.9 Module 9: Virtual Consultation Module

Purpose: Enable remote appointments and consultations.

Features:

- Schedule and manage virtual meetings
- Video call integration (e.g., Zoom API)
- Session notes and prescriptions upload
- Auto-reminder system for appointments

1.2.10 Module 10: Report & History Module

Purpose: Maintain health record history and generate reports.

Features:

- Generate weekly/monthly reports
- Exportable as PDF or CSV
- Timeline-based filters
- Medical summary for clinical visits

1.2.11 Module 11: Pharmacy & Lab Coordination Module

Purpose: Streamline communication with external healthcare services.

Features:

- Order prescriptions online
- Upload test results from diagnostic labs
- Lab notifications for pending/received reports
- Pharmacy delivery tracking (optional extension)

1.2.12 Module 12: Admin Panel

Purpose: Centralized control panel for platform maintenance and user management.

Features:

- View & manage all users
- Monitor AI model outputs/logs
- Suspend/delete suspicious accounts
- View feedback/complaints
- Generate usage analytics

1.3 Overview

This SRS document continues with detailed sections that outline all aspects of the system. It includes a comprehensive list of **system features**, covering both functional requirements (what the system should do, such as health data logging and alert generation) and non-functional requirements (how well the system performs, including security, performance, and usability). The **external interface requirements** describe how the platform communicates with software services like Firebase, APIs, and user devices. The **system design constraints** section highlights limitations such as required technologies and compliance with data standards. Visual **diagrams and use cases** will illustrate user interactions and system workflows for better understanding. Finally, the **testing and validation** section defines how the system will be evaluated to ensure it meets all requirements, including unit, integration, and user acceptance testing.

2 Overall Description:

Specifically designed for Pakistani patients with diabetes and hypertension, the project intends to provide an intelligent, AI-powered web platform for health monitoring. Since many patients lack timely access to medical advice and the prevalence of these chronic diseases is on the rise, the suggested approach aims to fill the gaps in existing health management methods.

The platform will give customers a digitized, centralized way to record important health information like blood pressure and blood sugar levels. Advanced machine learning algorithms such as LSTM, SVM, and Logistic Regression will subsequently be used to analyse this data in order to assess risks, predict future health trends, and provide individualized health warnings. It is an all-in-one solution for patient monitoring and assistance thanks to features like real-time notifications, automatic insulin dose recommendations, visual health statistics, and remote doctor consultations.

User administration, health data entry, AI predictions, emergency alerting, dashboards, professional panels (for physicians and caregivers), and pharmacy/lab coordination are just a few of the modules that make up the system. All of these modules work together seamlessly.

Notably, the solution prioritizes affordability, usability, and accessibility with the goal of enhancing the healthcare experience for individuals from all socioeconomic backgrounds. Despite being entirely digital, the application does not provide in-person consultations or physical therapy.

In conclusion, the initiative represents a step toward improving patient, caregiver, and healthcare provider cooperation for better health outcomes and digitizing chronic care management in Pakistan with predictive AI technology.

2.1 Product Perspective:

The suggested product is a web-based, stand-alone health monitoring system that incorporates artificial intelligence to help people with hypertension and diabetes. Although it is not a component of any current system, its modular architecture enables future connection with hospital databases, mobile apps, and other healthcare systems.

The platform would act as a centralized digital assistant, making it easier for patients, physicians, caregivers, and healthcare service providers to enter, track, analyze, and communicate health data. Although the system was created from the ground up, it is influenced by pre-existing digital health solutions such as One Drop, Livongo, and Omada Health. In low-resource environments like Pakistan, it seeks to overcome their shortcomings, particularly with regard to price, accessibility, and real-time response.

- The system is made up of many components that are linked yet operate separately:
- Role-based interactions and safe access are provided via a user management system.
- An AI prediction engine that forecasts health indicators using LSTM, SVM, and logistic regression.
- Alert System to notify pertinent parties in a timely manner.
- A virtual consultation module to let patients and doctors communicate online.
- Integration of Pharmacy and Lab to enhance treatment and diagnostic coordination.

From the perspective of the user, the product will work as an independent health monitoring tool that can be accessed via any contemporary web browser. Based on predictive analytics, it is intended to reduce human overhead, enhance decision-making, and offer prompt interventions.

It is anticipated that the product will be scalable and capable of:

1. Make a smartphone version available. (Optional)
2. Encourage more illnesses.
3. In the future, establish connections with national electronic health record (EHR) systems.

2.2 User Classes and Characteristics:

Our system is made to accommodate many user classes, each of which has different roles, duties, and technical proficiency levels. To guarantee smooth healthcare monitoring, predictive analysis, and prompt response, these users engage with the platform in a variety of ways.

UC-1: Patients (Primary Users) :

- **Description:** The platform is used by people with diabetes or hypertension to record vital signs, track their health, and get AI-based suggestions.
- **Technical Expertise:** Fundamental abilities to use a computer or smartphone.
- **Responsibilities:** Entering health information, answering notifications, looking over patterns and recommendations, and participating in online discussions.
- **Usage Frequency:** Every day or several times a week.
- **Key Requirements:** Medication reminders, graphic health summaries, linguistic support, and an easy-to-use interface.
- **Training Needs:** Data input, chart interpretation, and system alert response are examples of basic onboarding.

UC-2: Doctors / Healthcare Professionals:

- **Description:** Licensed physicians who provide medical advice, authorize AI-generated recommendations, and keep an eye on patients' medical data.
- **Technical Expertise:** Proficiency with medical applications and computers ranges from intermediate to advanced.
- **Responsibilities:** Examine medical records, authorize or adjust insulin doses, write prescriptions, and have online consultations.
- **Usage Frequency:** Regular, based on the number of patients.
- **Key Requirements:** Comprehensive dashboards, communication tools, connection with virtual meetings, and access to patient histories
- **Training Needs:** Navigating the system dashboard, comprehending AI-based recommendations, and analysing reports.

UC-3: Caregivers / Family Members:

- **Description:** Dependable family members or caretakers keeping an eye on the patient's health and helping with care.
- **Technical Expertise:** Basic knowledge of smartphones or computers.
- **Responsibilities:** Track health trends, keep an eye on alarms, work with physicians, and encourage drug compliance.
- **Usage Frequency:** Periodic; elevated amid changes in health.
- **Key Requirements:** Real-time alarm messages, read-only access to medical records, and channels of contact with patients and doctors.
- **Training Needs:** Dashboard overview, alarm reaction, and use of communication tools.

UC-4: System Administrators:

- **Description:** IT personnel are in charge of backend management, guaranteeing system security, performance, and availability.
- **Technical Expertise:** advanced knowledge of database administration, cloud infrastructure, web development, and AI integration.
- **Responsibilities:** System configuration, deployment of AI models, log monitoring, data privacy, and compliance.
- **Usage Frequency:** step of setup and any necessary upkeep or troubleshooting.

- **Key Requirements:** monitoring logs, an administrative control interface, and statistics on AI performance and system utilization.
- **Training Needs:** Security, compliance procedures, ML pipeline management, and system design.

UC-5: Pharmacists / Lab Technicians (Optional Integration Users):

- **Description:** outside medical partners engaged in lab testing or prescription fulfillment.
- **Technical Expertise:** Basic to intermediate digital systems understanding.
- **Responsibilities:** basic to intermediate knowledge of digital systems.
- **Usage Frequency:** Occasionally, based on user requirements
- **Key Requirements:** tools for status updates, upload portals, and notification systems.
- **Training Needs:** Notification handling, document management, and portal access training.

2.3 Operating Environment

The AI-based Health Monitoring System is made to work in a web-based, cloud-supported environment and be accessed from a range of devices with dependable internet access. The system is scalable, secure, and performance-optimized to manage real-time predictive analytics and sensitive medical data.

2.3.1 Hardware Platform:

- **OE-1:** The system must run on physical servers with an x86-64 architecture (at least an Intel Core i5 or Ryzen 5) or on servers hosted in the cloud.
- **OE-2:** To accommodate many users and AI processing, the system must have at least 8GB of RAM for development and 16GB for deployment.
- **OE-3:** For machine learning model training, GPU-enabled environments (e.g., NVIDIA RTX 2060 or higher) shall be used.
- **OE-4:** A minimum of 256GB SSD storage shall be used for application deployment; extended cloud storage (S3 or equivalent) will handle health logs and reports.

2.3.2 Operating System:

- **OE-5:** For a stable server deployment, the backend must operate on Ubuntu 20.04 LTS or a later Linux version.
- **OE-6:** macOS (10.15 or higher) or Windows 10/11 Professional may be used in development and local testing environments.
- **OE-7:** For consistency in the environment and deployment convenience, Docker containerization will be utilized.

2.3.3 Network Environment:

- **OE-8:** The system must function across secure HTTPS connections on firewall-protected public or private networks.

- **OE-9:** Real-time features like dashboard statistics and virtual consultations require a minimum connection bandwidth of 5 Mbps.
- **OE-10:** The system will use secure TCP/IP protocols to connect to external APIs (video services, email/SMS gateways).
- **OE-11:** In order for the platform to perform essential features like data synchronization, AI model usage, and alarm alerts, internet access is required.

2.3.4 Web Browser Support:

- **OE-12:** Google Chrome version 90 and higher are required to run the system dashboard properly.
- **OE-13:** Mozilla Firefox version 88 and higher must be supported by the system.
- **OE-14:** Version 90 or later of Microsoft Edge (which is based on Chrome) must be compatible with the platform.
- **OE-15:** For smartphone browser access, the user interface will be responsive and mobile-friendly.

2.3.5 Database Environment:

- **OE-16:** For structured data such user records, login histories, and medical logs, the system must employ PostgreSQL (version 13 or higher).
- **OE-17:** Semi-structured data, including unstructured feedback, AI findings, and dynamic health trends, will be stored in MongoDB (version 5.0 or higher).
- **OE-18:** All databases must be housed on cloud services that are safe, encrypted, and have automatic backups.

2.4 Design and Implementation Constraints:

To guarantee compatibility, performance, scalability, and the safe handling of private medical data, the AI-powered Health Monitoring System's design and development must follow certain technological and environmental requirements.

2.4.1 Operating System:

- **C-1:** For server-side processing and backend deployment, the system must run on Ubuntu 20.04 LTS or a later Linux version.
- **C-2:** Windows 10/11 Professional for local testing, debugging, and development environments must be supported by the system.
- **C-3:** To guarantee consistent environments, simplicity of deployment, and support for microservice design, the system must be compatible with Docker containerization.

2.4.2 Network Environment:

- C-4: In order to perform essential features like real-time notifications, virtual consultations, and cloud database access, the system must have active internet connectivity and run across secure public and private networks.
- C-5: The system will use HTTPS/TCP/IP protocols to connect to other services (such as video conferencing platforms and email/SMS APIs).
- C-6: To safeguard patient privacy and guarantee adherence to data protection laws, end-to-end encryption mechanisms must be utilized for all data transfer.

2.4.3 Web Browser Support:

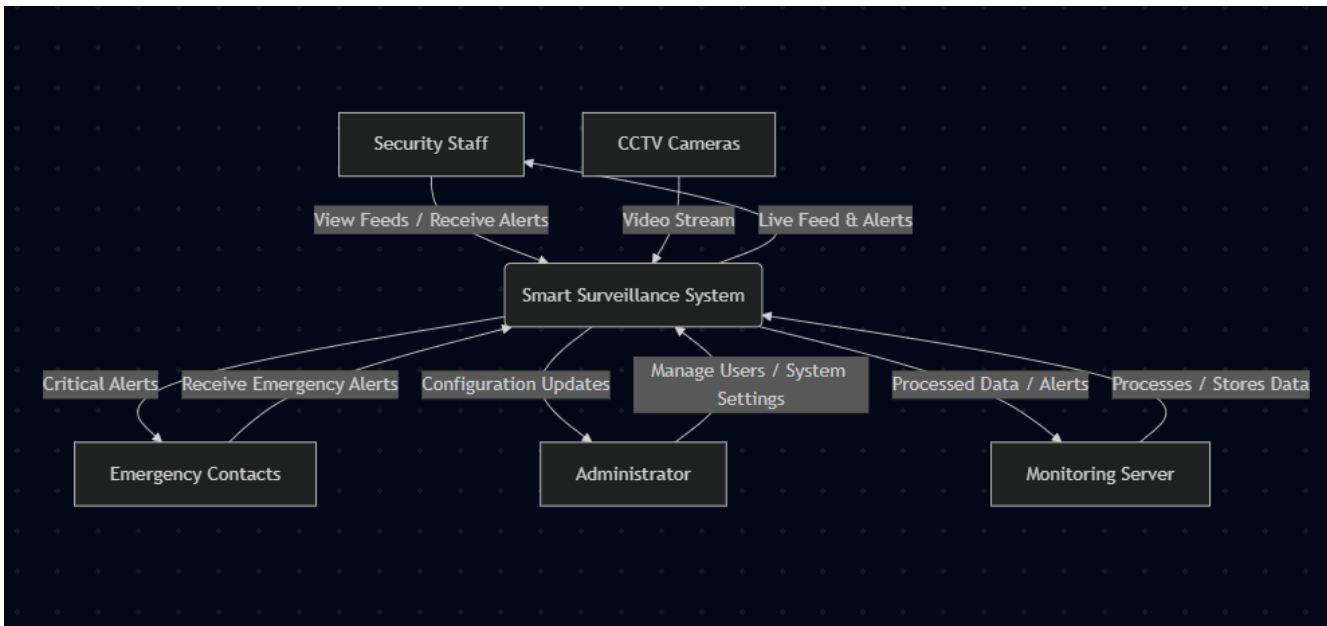
- C-7: Google Chrome version 90 and higher are required to operate the system dashboard properly.
- C-8: Versions 88 and higher of Mozilla Firefox must be supported by the system.
- C-9: Version 90 and later of Microsoft Edge (which is based on Chrome) must be compatible with the platform.
- C-10: For easy access on smartphones and tablets, the user interface has to be responsive and mobile-friendly.

2.4.4 Database Environment:

- C-11: To manage relational data, including users, logs, and appointment calendars, the system must employ PostgreSQL version 13 or later.
- C-12: MongoDB version 5.0 or later must be supported by the system in order to store semi-structured or unstructured data, such as feedback, health trends, and AI outputs.
- C-13: Every database must be housed on a secure cloud platform with integrated encryption and automatic backups, such as AWS RDS or MongoDB Atlas.

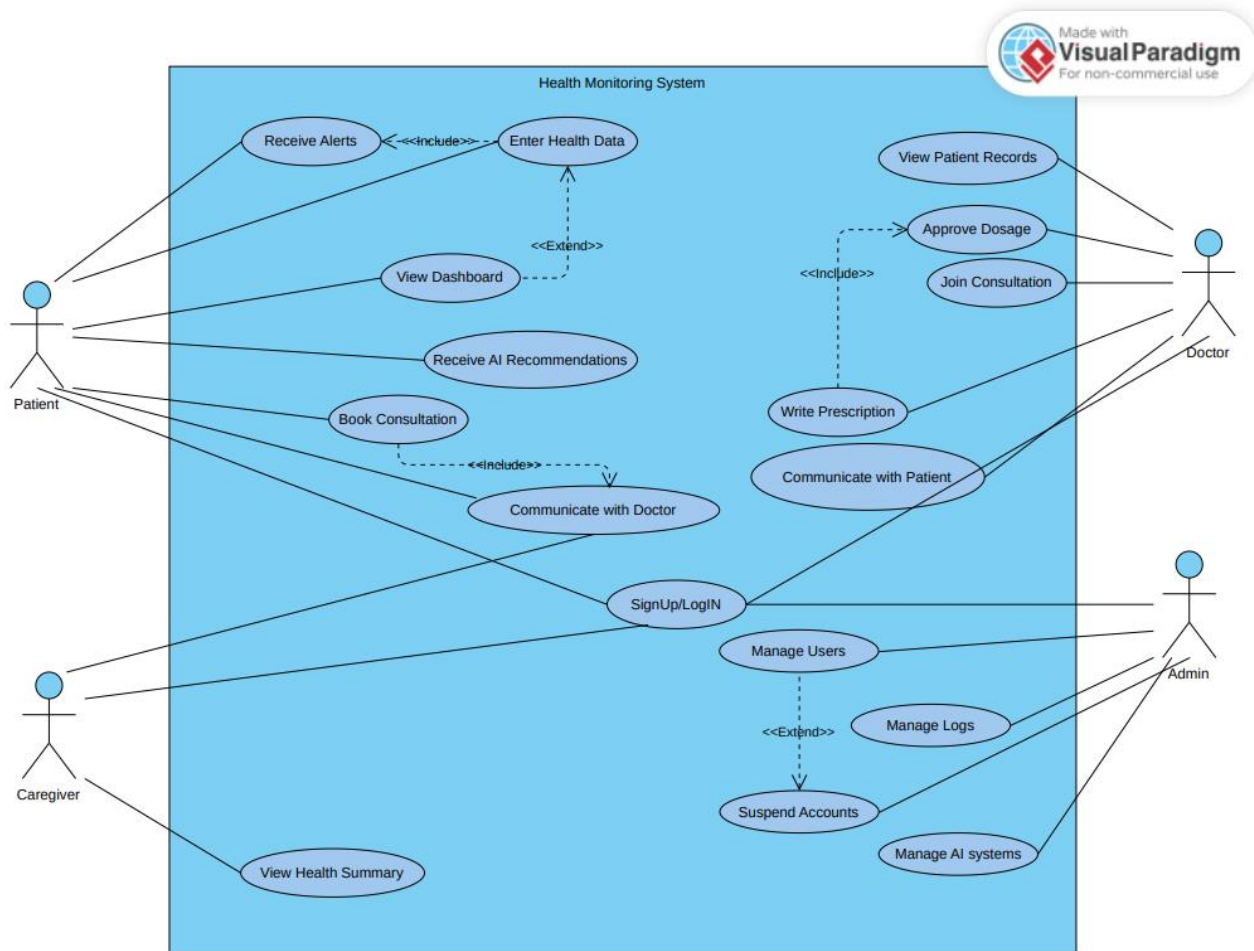
2.4.5 AI & ML Constraints:

- C-14: AI modules must operate on hardware with GPU capability during training and be created using open-source libraries like TensorFlow, PyTorch, or Scikit-learn.
- C-15: The amount and caliber of health data gathered determines the accuracy of ML models, which may need constant fine-tuning and retraining.
- C-16: The system has to make sure that doctors evaluate AI-generated recommendations before making important decisions like insulin dose.



3 Requirement Identifying Technique:

We have choose the graphical, detailed use case and event response table for requirement identifying technique:



Event	System State	Response
Vital data submitted by patient	Health data entry module is active	Validate and store readings (BP, sugar, etc.) in user profile.
Abnormal vital signs detected	AI model is active	Generate alert and send notifications to patient, doctor, and caregiver.
User logs in	User authentication module is active	Redirect to role-based dashboard (patient, doctor, caregiver, admin).
AI suggests insulin dosage change	AI prediction module has analysed trends	Notify doctor for approval with detailed reasoning and chart visual.

Doctor approves insulin suggestion	Doctor dashboard is active	Apply change and update patient's dosage plan in records.
Medication time reached	Notification scheduler is active	Send push/SMS/email reminders to patient.
Emergency condition forecasted	Predictive analytics model detects risk	Send immediate alerts to all linked users (doctor, caregiver, patient).
Virtual consultation scheduled	Appointment system is active	Generate session link and send reminders to all participants.
Lab report uploaded	Lab coordination module is active	Attach report to patient profile and notify doctor.
Patient requests report	Report generation module is active	Generate and export report in PDF/CSV format for selected time period.
Admin logs in	Admin dashboard is authenticated	Display full platform usage, AI logs, feedback, and system health summary.
AI model retrained	New training data available	Update model and adjust thresholds based on feedback and accuracy review

3.1 Detailed Use Case:

3.1.1 UC-1 - User Registration and Role-based Login

Use Case ID	UC-1
Use Case Name	User Registration and Role-based Login
Actors	Patient, Doctor, Caregiver
Description	Allows different users to create accounts and access personalized dashboards.
Trigger	User opens the registration/login page.
Preconditions	Internet connection; access to web platform.
Post conditions	Role-based dashboard is shown after login.

Normal Flow	1. User opens the registration/login page. 2. User selects role (Patient/Doctor/Caregiver). 3. User fills in required details. 4. Credentials are verified. 5. Role-based dashboard is shown after login.
Alternative Flows	1.1 Incorrect credentials: Show error. 1.2 Forgot password: Provide password recovery link.
Exceptions	1.0.E1 Server downtime: Notify user and retry later.
Business Rules	All roles must have separate dashboard permissions.
Assumptions	All users have valid email accounts for verification.

3.1.2 UC-2 - Manual Health Data Entry

Use Case ID	UC-2
Use Case Name	Manual Health Data Entry
Actors	Patient
Description	Patients enter health metrics like blood sugar, blood pressure, etc.
Trigger	User navigates to 'Health Data Entry'.
Preconditions	User is logged in.
Post conditions	Data is logged with a timestamp.
Normal Flow	1. User navigates to 'Health Data Entry'. 2. Inputs vitals (blood sugar, BP, heart rate). 3. Saves entry; data is logged with a timestamp.
Alternative Flows	1.1 User skips input: Prompt for completion.
Exceptions	1.0.E1 Internet loss during save: Retry on reconnect.
Business Rules	Health values must be within realistic bounds.
Assumptions	User measures vitals using correct tools.

3.1.3 UC-3 - AI-based Prediction & Trend Analysis

Use Case ID	UC-3
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Use Case Name	AI-based Prediction & Trend Analysis
Actors	System (AI), Patient, Doctor
Description	System predicts trends using ML models.
Trigger	New data is added or accessed.
Preconditions	Historical health data is available.
Post conditions	Trends are shown on dashboard.
Normal Flow	1. System fetches recent vitals. 2. LSTM forecasts glucose levels. 3. SVM/Logistic Regression predicts BP risk. 4. Trends are shown on dashboard.
Alternative Flows	1.1 Not enough data: Skip prediction and inform user.
Exceptions	1.0.E1 AI module error: Log issue and alert support.
Business Rules	AI must update predictions at least weekly.
Assumptions	AI model is trained and integrated with backend.

3.1.4 UC-4 - Insulin Dosage Recommendations

Use Case ID	UC-4
Use Case Name	Insulin Dosage Recommendations
Actors	AI System, Patient, Doctor
Description	Suggests insulin adjustments based on AI predictions.
Trigger	New diabetic data entry is available.
Preconditions	User has diabetes data logged.
Post conditions	Notification sent to patient.
Normal Flow	1. System analyzes glucose trends. 2. AI suggests dosage adjustment (increase/decrease). 3. Doctor approves/rejects suggestion. 4. Notification sent to patient.
Alternative Flows	1.1 Doctor disagrees: Overrides AI suggestion.
Exceptions	1.0.E1 AI fails to suggest dose: Notify doctor only.
Business Rules	Only doctors can approve insulin changes.
Assumptions	Patient data is recent and accurate.

3.1.5 UC-5 - Real-time Alerts & Notifications

Use Case ID	UC-5
Use Case Name	Real-time Alerts & Notifications
Actors	System, Patient, Doctor, Caregiver
Description	Notifies users about abnormal vitals or missed medications.
Trigger	AI detects abnormal data.
Preconditions	Health thresholds configured.
Post conditions	Alert sent to all concerned.
Normal Flow	1. AI checks input against health thresholds. 2. If risk is detected, send alerts via SMS/email/app. 3. Emergency contacts are notified if critical.
Alternative Flows	1.1 Patient disables alerts: Store log silently.
Exceptions	1.0.E1 SMS gateway fails: Use email as fallback.
Business Rules	High-risk alerts must trigger all notification channels.
Assumptions	Emergency contacts are registered and reachable.

3.1.6 UC-6 - Data Visualization

Use Case ID	UC-6
Use Case Name	Data Visualization
Actors	Patient, Doctor
Description	Displays health trends in visual formats.
Trigger	User accesses dashboard.
Preconditions	Data entries must exist.
Post conditions	Graphical display rendered.
Normal Flow	1. User selects desired time range. 2. System shows line/bar/pie graphs for vitals. 3. Risk status (Low/Med/High) indicated.
Alternative Flows	1.1 No data available: Display default message.
Exceptions	1.0.E1 Graph rendering fails: Show raw data table.

Business Rules	Visuals should be generated in under 2 seconds.
Assumptions	Graphs library is fully functional in supported browsers.

3.1.7 UC-7 - Doctor Dashboard & Recommendations

Use Case ID	UC-7
Use Case Name	Doctor Dashboard & Recommendations
Actors	Doctor
Description	Doctors access assigned patient data, approve AI suggestions, and issue prescriptions.
Trigger	Doctor logs into dashboard.
Preconditions	Doctor is authenticated.
Post conditions	Patient updates and recommendations issued.
Normal Flow	1. Doctor logs in and views assigned patients. 2. Reviews vitals, AI insights. 3. Approves/rejects insulin recommendations. 4. Uploads prescriptions or schedules appointments.
Alternative Flows	1.1 Doctor reschedules appointment or action.
Exceptions	1.0.E1 Patient data fails to load: Retry or report error.
Business Rules	Doctors can only view assigned patients.
Assumptions	Doctor's panel is updated in real time.

3.1.8 UC-8 - Caregiver Monitoring Panel

Use Case ID	UC-8
Use Case Name	Caregiver Monitoring Panel
Actors	Caregiver
Description	Caregivers monitor patient vitals and receive emergency alerts.
Trigger	Caregiver logs into dashboard.
Preconditions	Patient must add caregiver to their profile.
Post conditions	Vitals and alerts displayed.

Normal Flow	1. Caregiver logs in. 2. Views patient's vitals (read-only). 3. Receives emergency alerts.
Alternative Flows	1.1 Caregiver not linked to any patient: Show error.
Exceptions	1.0.E1 Alert delivery fails: Retry later.
Business Rules	Caregiver access is limited to read-only.
Assumptions	Caregiver has consent from patient.

3.1.9 UC-9 - Virtual Consultation

Use Case ID	UC-9
Use Case Name	Virtual Consultation
Actors	Patient, Doctor
Description	Schedule and conduct remote health sessions.
Trigger	Patient requests appointment.
Preconditions	Internet, Zoom API integration.
Post conditions	Session conducted and notes uploaded.
Normal Flow	1. Patient requests appointment. 2. Doctor confirms time. 3. Zoom link sent to both. 4. Session conducted; doctor uploads notes.
Alternative Flows	1.1 Doctor proposes alternate time: Patient confirms.
Exceptions	1.0.E1 Video call drops: Notify and reschedule.
Business Rules	Each consultation limited to 30 minutes.
Assumptions	Both parties have webcam and mic access.

➤ UC-10 - Report Generation

Use Case ID	UC-10
Use Case Name	Report Generation
Actors	Patient, Doctor
Description	Automatically generate downloadable health reports.
Trigger	User selects report timeframe.
Preconditions	Historical data must be present.
Post conditions	Report downloaded.

Normal Flow	1. User selects timeframe (weekly/monthly). 2. System compiles data. 3. User downloads as PDF or CSV.
Alternative Flows	1.1 No data in selected period: Show info message.
Exceptions	1.0.E1 PDF/CSV generation fails: Retry or export partial data.
Business Rules	Reports must include charts and summaries.
Assumptions	Data is cleaned and pre-processed.

3.1.10 UC-11 - Pharmacy & Lab Coordination

Use Case ID	UC-11
Use Case Name	Pharmacy & Lab Coordination
Actors	Patient, Doctor, Lab, Pharmacy
Description	Manage prescriptions, lab uploads, and deliveries.
Trigger	Prescription or test result uploaded.
Preconditions	Prescription generated or lab test done.
Post conditions	Interactions logged and tracked.
Normal Flow	1. Doctor uploads prescription. 2. Patient places order or sends to pharmacy. 3. Patient uploads test results from labs. 4. System logs all interactions.
Alternative Flows	1.1 Patient uploads expired report: Show warning.
Exceptions	1.0.E1 File upload fails: Prompt user to retry.
Business Rules	Only certified labs and pharmacies can be linked.
Assumptions	External APIs are available and authenticated.

➤ UC-12 - Admin Management

Use Case ID	UC-12
Use Case Name	Admin Management
Actors	Admin
Description	Manage users, monitor logs, and handle complaints.

Trigger	Admin logs in.
Preconditions	Admin access granted.
Post conditions	Platform maintained and issues resolved.
Normal Flow	1. Admin logs in. 2. Views user accounts, AI logs. 3. Takes action against flagged accounts. 4. Generates platform usage reports.
Alternative Flows	1.1 Admin flags a suspicious user for review.
Exceptions	1.0.E1 Server analytics tool error: Fallback to log file.
Business Rules	Admin actions must be audit-logged.
Assumptions	Admins have secure elevated access.

4 Functional Requirements

DailyMed Ai Have the Following Functional Requirements:

- The system shall stream **live camera feed** on the web application for authorized security staff.
- The system shall **detect motion or unauthorized access** using real-time video processing.
- The system shall **generate alerts** when suspicious activity is detected based on predefined rules (e.g., motion after office hours).
- The system shall **send notifications** (via SMS, email, or in-app) to the security staff upon detection of any suspicious activity.
- The system shall **log all detected events** with timestamps and video snapshot references for audit purposes.
- The system shall allow **users to review past video footage** based on time filters.
- The system shall enable **role-based access** — administrators can manage users, security staff can view feeds and alerts, while auditors can access logs
- The system shall allow security staff to **acknowledge and respond** to alerts through the dashboard.
- The system shall support **multi-camera integration**, showing multiple feeds on a single dashboard.
- The system shall maintain a **centralized alert history** with filtering by date, severity, and response status.

4.1 Functional Requirement X

4.1.1 FR-01: User Registration & Login

Identifier	FR-01
Title	User Registration & Login
Requirement	The system shall allow users (patients, doctors, caregivers) to register, log in, and authenticate using encrypted credentials.
Source	System Specification
Rationale	Secure access ensures role-based functionalities and data protection.
Business Rule (if required)	Users must use valid email and secure password for registration.
Dependencies	None
Priority	High

4.1.2 FR-02: Profile Management

Identifier	FR-02
Title	Profile Management
Requirement	Users shall be able to update or delete their profile information.
Source	System Specification
Rationale	Allows users to manage their data and keep it up to date.
Business Rule (if required)	Only authenticated users can edit their own profiles.
Dependencies	FR-01
Priority	Medium

4.1.3 FR-03: Health Data Entry

Identifier	FR-03
Title	Health Data Entry
Requirement	The system shall allow users to input and log vital health data (blood pressure, sugar, etc.) with time stamps.
Source	System Specification
Rationale	Supports analysis and tracking of health metrics over time.
Business Rule (if required)	Daily entry prompts will be triggered if data is not logged.

Dependencies	FR-01
Priority	High

4.1.4 FR-04: File Attachment

Identifier	FR-04
Title	File Attachment
Requirement	Users shall be able to upload lab results and health documents.
Source	System Specification
Rationale	Enables comprehensive health tracking and doctor verification.
Business Rule (if required)	Accepted formats include PDF, JPG, and PNG.
Dependencies	FR-03
Priority	Medium

4.1.5 FR-05: Daily Reminders

Identifier	FR-05
Title	Daily Reminders
Requirement	The system shall send reminders to users to log their daily health vitals.
Source	System Specification
Rationale	Ensures regular health tracking to enable accurate predictions.
Business Rule (if required)	Reminder sent via SMS, email, or app push.
Dependencies	FR-03
Priority	High

4.1.6 FR-06: AI Prediction (Blood Sugar)

Identifier	FR-06
Title	AI Prediction (Blood Sugar)
Requirement	System shall use LSTM models to forecast future blood sugar levels.
Source	System Specification

Rationale	Provides early warnings to prevent hypo/hyperglycemia.
Business Rule (if required)	Data must be from at least 7 days to generate forecast.
Dependencies	FR-03
Priority	High

4.1.7 FR-07: AI Risk Analysis (BP)

Identifier	FR-07
Title	AI Risk Analysis (BP)
Requirement	System shall use SVM or Logistic Regression to analyze hypertension risk.
Source	System Specification
Rationale	Supports proactive care for high BP cases.
Business Rule (if required)	Weekly analysis per user record.
Dependencies	FR-03
Priority	High

4.1.8 FR-08: Dosage Recommendation

Identifier	FR-08
Title	Dosage Recommendation
Requirement	System shall suggest insulin dosage based on AI analysis.
Source	System Specification
Rationale	Helps manage diabetes efficiently and personally.
Business Rule (if required)	Requires doctor approval before application.
Dependencies	FR-06
Priority	High

4.1.9 FR-09: Doctor Approval

Identifier	FR-09
Title	Doctor Approval

Requirement	Doctors shall be able to review and approve/reject AI-generated dosage suggestions.
Source	System Specification
Rationale	Ensures medical oversight on AI actions.
Business Rule (if required)	Approval logs must be stored.
Dependencies	FR-08
Priority	Medium

4.1.10 FR-10: Emergency Alerting

Identifier	FR-10
Title	Emergency Alerting
Requirement	System shall notify emergency contacts if critical health stats are logged.
Source	System Specification
Rationale	Ensures quick response during patient crises.
Business Rule (if required)	Alert priority escalated based on risk.
Dependencies	FR-06, FR-07
Priority	High

4.1.11 FR-11: Medication Reminders

Identifier	FR-11
Title	Medication Reminders
Requirement	System shall send reminders to users to take medications.
Source	System Specification
Rationale	Improves adherence to prescribed treatments.
Business Rule (if required)	Customizable reminder schedule per user.
Dependencies	FR-01
Priority	Medium

4.1.12 FR-12: Dashboard Visualizations

Identifier	FR-12
Title	Dashboard Visualizations

Requirement	System shall display trends and summaries of vitals using charts.
Source	System Specification
Rationale	Enables users and doctors to quickly assess health trends.
Business Rule (if required)	Supports line, pie, and bar graphs.
Dependencies	FR-03
Priority	Medium

4.1.13 FR-13: Doctor Panel

Identifier	FR-13
Title	Doctor Panel
Requirement	Doctors shall access assigned patient profiles, approve AI recommendations, and prescribe.
Source	System Specification
Rationale	Improves doctor-patient coordination and decision-making.
Business Rule (if required)	Access restricted to assigned patients only.
Dependencies	FR-01
Priority	High

4.1.14 FR-14: Caregiver Access Panel

Identifier	FR-14
Title	Caregiver Access Panel
Requirement	Caregivers shall be able to view real-time alerts and read-only health data.
Source	System Specification
Rationale	Keeps caregivers informed without altering records.
Business Rule (if required)	No modification rights for caregivers.
Dependencies	FR-01
Priority	Medium

4.1.15 FR-15: Virtual Consultation

Identifier	FR-15
Title	Virtual Consultation
Requirement	System shall provide scheduling and video call support for remote consultations.
Source	System Specification
Rationale	Enhances accessibility and continuity of care.
Business Rule (if required)	Integrated with Zoom or similar APIs.
Dependencies	FR-01
Priority	High

4.1.16 FR-16: Report Generation

Identifier	FR-16
Title	Report Generation
Requirement	System shall generate and export PDF/CSV reports based on user health logs.
Source	System Specification
Rationale	Enables sharing of summaries with clinics or doctors.
Business Rule (if required)	Reports generated weekly or monthly.
Dependencies	FR-03
Priority	Medium

4.1.17 FR-17: Pharmacy Integration

Identifier	FR-17
Title	Pharmacy Integration
Requirement	Users shall be able to order prescriptions online from pharmacies.
Source	System Specification
Rationale	Simplifies medication acquisition.
Business Rule (if required)	Pharmacy verification required before dispatch.
Dependencies	FR-13
Priority	Medium

4.1.18 FR-18: Lab Integration

Identifier	FR-18
Title	Lab Integration
Requirement	Users shall upload lab results and receive result notifications.
Source	System Specification
Rationale	Maintains comprehensive health profiles.
Business Rule (if required)	Supports both manual and API-based upload.
Dependencies	FR-03
Priority	Medium

4.1.19 FR-19: Admin Management Panel

Identifier	FR-19
Title	Admin Management Panel
Requirement	Admin shall monitor users, AI logs, and manage suspicious activity.
Source	System Specification
Rationale	Ensures platform integrity and performance.
Business Rule (if required)	Admin can suspend/delete accounts.
Dependencies	FR-01
Priority	High

4.1.20 FR-20: Feedback & Complaints

Identifier	FR-20
Title	Feedback & Complaints
Requirement	Users shall submit feedback or complaints through a form.
Source	System Specification
Rationale	Improves system quality based on user insights.
Business Rule (if required)	Anonymous and named feedback supported.
Dependencies	FR-01
Priority	Low

4.1.21 FR-21: Account Moderation

Identifier	FR-21
Title	Account Moderation
Requirement	Admin shall manage user bans or suspensions for policy violations.
Source	System Specification
Rationale	Maintains a safe and secure environment.
Business Rule (if required)	Requires internal report verification before action.
Dependencies	FR-19
Priority	High

5 Non-Functional Requirements

DailyMed Ai Have the Following Non-Functional Requirements:

- **NFR-1:** The system shall maintain 99.5% uptime for video feed availability
- **NFR-2:** The alert generation delay after incident detection shall not exceed 2 seconds.
- **NFR-3:** The system shall use SSL encryption to secure live feeds and stored video data.
- **NFR-4:** The system shall be compatible with modern web browsers including Chrome, Firefox, and Edge.
- **NFR-5:** The platform shall support scaling up to 100 concurrent users without performance degradation.
- **NFR-6:** The user interface shall be designed with responsive layouts to support access on desktop and mobile devices.
- **NFR-7:** The system shall retain all video and alert logs for a minimum of 90 days before automatic archival or deletion.

5.1.1 Reliability:

- **NFR-R1:** The system shall maintain a minimum uptime of 99.5%, ensuring consistent surveillance coverage.
- **NFR-R2:** The system shall automatically reconnect to video streams within 5 seconds of a temporary disconnection.

- **NFR-R3:** All alerts and events must be backed up to cloud storage every 24 hours to prevent data loss.

5.1.2 Usability:

- **NFR-U1:** The system shall offer an intuitive and user-friendly interface for non-technical users such as security staff.
- **NFR-U2:** The web dashboard shall support visual indicators (e.g., red/green status lights, icons) for camera and alert statuses.
- **NFR-U3:** Users shall be able to learn to operate the system within 30 minutes using an embedded help guide and tooltips.
- **NFR-U4:** The system shall provide multi-language support, starting with English and Urdu.

5.1.3 Performance:

- **NFR-P1:** The system shall display live camera feed with less than 2 seconds delay under standard network conditions.
- **NFR-P2:** Alert processing from video stream to notification dispatch shall occur in under 3 seconds.
- **NFR-P3:** The platform shall support up to 50 simultaneous video streams with no more than 10% CPU utilization on a dedicated server.
- **NFR-P4:** The dashboard shall load completely in under 5 seconds for standard broadband users.

5.1.4 Security:

- **NFR-S1:** All video streams and data transmissions shall be encrypted using SSL/TLS protocols.
- **NFR-S2:** The system shall implement role-based access control to restrict access based on user type (admin, security staff, auditor).
- **NFR-S3:** Users shall be required to authenticate using multi-factor authentication (MFA) during login.
- **NFR-S4:** The system shall log all user activity (e.g., login, alert acknowledgment) and retain it for a minimum of 90 days.
- **NFR-S5:** The application shall be protected against common threats such as SQL injection, XSS, and brute-force attacks.

6 User Interfaces Requirements

- **UI/UX Standards-** The system will follow Material Design Guidelines to maintain consistency in the application's layout and appearance. The design will also prioritize mobile first responsiveness

to adapt to various screen sizes.

- **Font, Icon and Label Standards-** “Roboto” font will be primarily used. Icons from Material Icons will be used. Action based labels will be used for various actions.
- **Color Scheme-** Primary: #2E86C1 (Blue – for actions)
Secondary: #AED6F1 (Light Blue – highlights)
Error: #E74C3C (Red – alerts/warnings)
Success: #27AE60 (Green – success states)
- **Common Layout and Navigation Elements-** The header will include a fixed navigation bar will include a logo, user profile icon, notification bell icon and a help icon. The footer will include contact information, version and links to policies and terms.
- **Standard Controls and Field Behavior-** Form fields will include date pickers, sliders and dropdown menus. Inline validation and real-time feedback will be included.
- **Screen Resolution and Layout Constraints-** Minimum supported resolution will be 1366x768 and other resolutions will be optimized accordingly.
- **Localization Layout Standards-** All text content will be externalized in JSON locale files to allow easy translation.
- **Message Display Conventions-** Error messages will be displayed in red with respective icon. Success messages will be shown in green with respective icon and notifications will follow uniform structure

6.1 Software interfaces

SI-1: MongoDB 7.0

SI-1.1: The system shall store and retrieve patient health data, medication logs, and AI prediction results in the MongoDB database via a secure backend API.

SI-2: Zoom API (video call conferencing)

SI-1.1: The system shall integrate with zoom API to conduct online consultations of doctors with patients.

SI-3: Email/SMS Gateway

SI-1.1: The system shall utilize third party APIs to send medication reminders, alerts and appointment notifications.

SI-4: Machine Learning Libraries

SI-1.1: The system shall use machine learning libraries to predict and process necessary data using LSTM, SVM and Logistic Regression models.

6.2 Communications interfaces

CI-1: The system shall send an email or text message (based on user account settings) to notify patients, caregivers or doctors about consultations, dosage and necessary reminders or alerts.

CI-2: The system shall operate over HTTPS using standard TCP/IP protocols to ensure secure communication between the frontend and the backend.

CI-3: The system shall use real-time web technologies for live health alerts and status updates on dashboards

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